1 ECE 215A Notes

1.1 MOS Transistor

MOSFET current relation obtain through depletion region

$$\begin{aligned} V_{th} =& \Phi_{MS} + 2\Phi_F + \frac{Q_{dep}}{C_o x} \\ Q_d =& WC_{ox}(V_{GS} - V_{th} - V(x)) \\ I_D =& -WC_{ox}(V_{GS} - V_{th} - V(x))v \end{aligned}$$

Where v is

$$v = \mu \frac{dV(x)}{dx}$$

Let $\mu C_{ox} W/L = \beta$ for simplicity

$$I_D = \beta \left[(V_{GS} - V_{th})V_{DS} - \frac{1}{2}V_{DS}^2 \right]$$

$$I_{Dmax} = \frac{1}{2}\beta (V_{GS} - V_{th})^2$$

1.1.1 Trans-Conductance

At constant V_{DS}

$$\begin{aligned} g_m =& \frac{\delta I_D}{\delta V_{GS}} = \beta (V_{GS} - V_{th}) \\ =& \sqrt{2\beta I_D} = \frac{2I_D}{V_{GS} - V_{th}} \end{aligned}$$

or with CLM

$$g_m = \sqrt{\frac{2\beta I_D}{1 + \lambda V_{DS}}} \qquad V_{eff} \approx \sqrt{\frac{2I_D}{\beta}}$$

1.1.2 Secondary Effects

Body Effect

$$V_{TH} = V_{TH0} + \gamma(\sqrt{2\Phi_F + V_{SB}} - \sqrt{|2\Phi_F|})$$

$$\gamma = \frac{\sqrt{2q\varepsilon_{si} N_{sub}}}{C_{ox}}$$

Channel Length Modulation (Occurs after saturation) The longer L the smaller the λ

$$\begin{aligned} I_D \approx& \frac{1}{2}\beta (V_{GS} - V_{th})^2 (1 + \lambda V_{DS}) \\ r_o =& \frac{\delta V_{DS}}{\delta I_D} = \left[\frac{1}{2}\beta (V_{GS} - V_{th})^2 \lambda \right]^{-1} \\ \approx& \frac{1 + \lambda V_{DS}}{\lambda I_D} \approx \frac{1}{\lambda I_D} \end{aligned}$$

Subthreshold Conduction $\xi (> 1)$ is the nonideality factor

$$I_D = I_0 e^{V_{GS}/(\xi V_T)}$$

$$V(GS) - Vth = 2\xi V_T$$

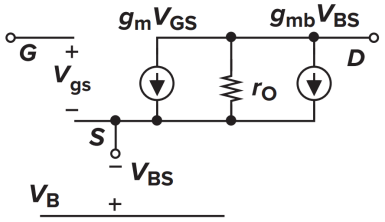
1.2 Transfer Function

Quality Factor and attenuation Factor

$$Q = \frac{\omega_0}{2\alpha} = \frac{\omega_0}{\omega_{-3dB}} \qquad \alpha = \frac{1}{\tau}$$

1.3 Small Signal Model

With Secondary Effects Included



Where the body effect coefficient is

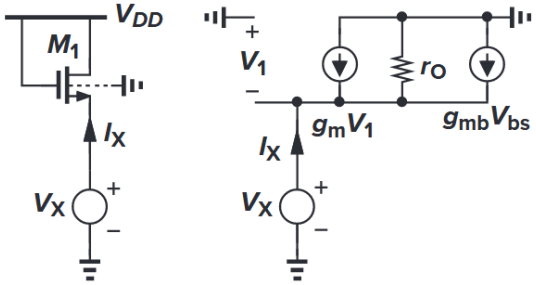
$$g_{mb} = \frac{\delta I_D}{\delta V_{BS}} = g_m \frac{\gamma}{2\sqrt{2\Phi_F + V_{SB}}} = \eta g_m$$

1.3.1 Gain

$A_v = G_m R_{out}$, Gm maybe find by Grounding Output and calculate I_{out}/V_{in}, R_{out} maybe find by Grounding Input and apply voltage V test at output V_{test}/I_{test}

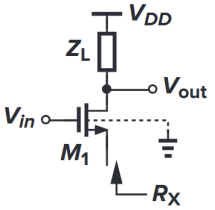
1.4 Impedance From Source

Diode Connected



$$\frac{V_X}{I_X} = \frac{1}{g_m + g_{mb}} || r_o \approx \frac{1}{g_m + g_{mb}}$$

Loaded, $\lambda = 0$

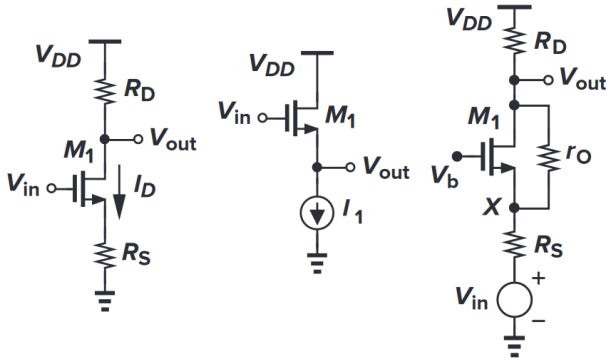


Impedance from below $\lambda = 0$ and $\lambda \neq 0$

$$\frac{V_X}{I_X} = \frac{1}{g'_m} \qquad \frac{V_X}{I_X} = (g'^{-1}_m || r_o) \left[1 + \frac{Z_L}{g_m^{-1} || r_o} \right]$$

Active load may provide

1.5 Single Stage Amplifier



CS (Degenerated), CD Source Follower, CG stage

1.5.1 Common Source stage with Degeneration

Without secondary effects

$$G_m = \frac{g_m}{1+g_m R_S} \qquad R_{out} = R_D$$

With secondary effects

$$G_m = \frac{g_m r_o}{R_S + [1+(g_m+g_{mb})R_S]r_o}$$

$$R_{out} = ([1+(g_m+g_{mb})r_o]R_S + r_o)||R_D$$

$$\approx [1+(g_m+g_{mb})R_S]r_o$$

For Both

$$R_{in} = \infty$$

1.5.2 Common Drain Source Follower

With secondary effects $R_{in} = \infty$

$$R_{out} = \frac{1}{g_m+g_{mb}}||r_o \qquad G_m = g_m$$

If the I_1 is not ideal (or replaced with a R_S)

$$\frac{V_{out}}{V_{in}} = \frac{g_m R_S}{1+g_m R_S}$$

1.5.3 Common Gate

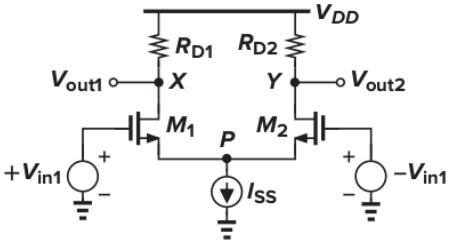
With secondary effects

$$G_m = \left[R_S + r_o || \frac{1}{g_{mb}+g_m}\right]^{-1}$$

$$R_{out} = [(1+(g_m+g_{mb})r_o)R_S + r_o]\,||R_D$$

$$R_{in} = \frac{r_o + R_D}{1+g'_m r_o} + R_S$$

1.6 Differential Pair



If not Symmstrical

$$A_D = \frac{\Delta V_{out}}{\Delta V_{in}} = -2(g_{m1}||g_{m2})R_D$$

With $\lambda \neq 0$

$$V_{out1} = -g_{m1}(R_{D1}||r_{o1})(V_{in1} - V_{in2})/2$$
$$V_{out2} = -g_{m2}(R_{D2}||r_{o2})(V_{in2} - V_{in1})/2$$

If completely symmetrical (symmetrical Circuit, you may adjust model to make them symmetrical, can be broken up into half-circuits to analyze)

$$A = -gm(R_D||r_o)$$

1.6.1 Common-Mode Response

$$A_{v,CM} = -\frac{R_D/2}{(g_{m1}+g_{m2})^{-1} + R_{SS}}$$

1.6.2 CMRR

Common Mode Rejection Ratio

$$CMRR = \left|\frac{A_{DM}}{A_{CM}}\right|$$

1.6.3 MOS Load

Diode Connected Load

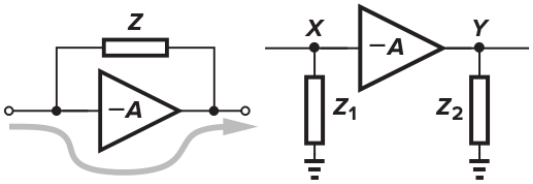
$$A_v = -g_{mN}(g_{mP}^{-1}||r_{ON}||r_{OP})$$

Current Source Load

$$A_v = -g_{mN}(r_{ON}||r_{OP})$$

2 Frequency Response

2.1 Miller's Theorem



$$A = \frac{V_Y}{V_X} \qquad Z_1 = \frac{Z}{1-A} \qquad Z_2 = \frac{Z}{1-A^{-1}}$$

2.2 Poles and Zeros

No inductor in this case so RC has pole $P = (RC)^{-1}$

$$H(s) = \frac{\Pi(s+z_i)}{\Pi(s+p_i)}$$

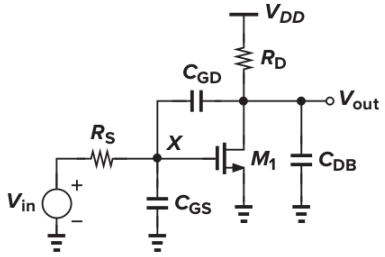
Always same amount of poles and zeros (just that some may be at infinity or zero), and there will always same poles as number of reactive elements (capacitor/inductor)

2.3 Poles of different Stages

2.3.1 Gain

$$A(s) = \frac{\text{DC Gain}}{(1 + s/\omega_{pin})(1 + s/\omega_{pout})}$$

2.3.2 Common Source



CS stage with all caps

$$\begin{cases} \omega_{pin} \approx [R_S(C_{gs} + C_{gd}(1 + g_m R_D))]^{-1} \\ \omega_{pout} \approx [R_D(C_{gd} + C_L)]^{-1} \end{cases}$$

Gain

$$A(s) = \frac{-g_m R_D}{(1 + s/\omega_{pin})(1 + s/\omega_{pout})}$$

Input Impedance

$$Z_{in} = \frac{1 + R_D(C_{GD} + C_{DB})s}{C_{GD}s(1 + g_m R_D + R_D C_{DB} s)}$$

2.3.3 Source Follower

$$\begin{cases} \omega_{pin} \approx [R_S C_{gd}]^{-1} \\ \omega_{pout} = \dfrac{g_m + R_L^{-1}}{C_{gs} + C_L} \end{cases}$$

Input Impedance

$$Z_{in} = \frac{1}{C_{GS}s} + \left(1 + \frac{gm}{C_{gs}s}\right)\frac{1}{g_{mb} + C_L s}$$

Output Impedance

$$Z_{out} = \frac{R_S C_{GS} s + 1}{g_m + C_{gs} s}$$

In this case equivalent to an inductance of

$$L = \frac{C_{GS}}{g_m}\left(R_S - \frac{1}{g_m}\right)$$

2.3.4 Common Gate

$$\begin{cases} \omega_{pin} \approx [C_{in}(g_m^{-1} || R_S)]^{-1} \\ \omega_{pout} = [R_D C_{out}]^{-1} \end{cases}$$

Input Impedance

$$Z_{in} \approx \frac{Z_L}{(g_m + g_{mb})r_o} + \frac{1}{g_m + g_{mb}}$$

2.3.5 Frequency Equations

From Equation

$$H(s) = \frac{f(s)}{s^2 + \frac{\omega_0}{Q}s + \omega^2}$$

With Feedback, β not a function of s

$$H_{FB}(s) = \frac{H(s)}{1 + \beta H(s)},\ H(s) = \frac{A_{DC}}{(\frac{s}{\omega_{p1}} + 1)(\frac{s}{\omega_{p2}} + 1)}$$
$$= \frac{A_{DC}\omega_{p1}\omega_{p2}}{s^2 + s(\omega_{p1} + \omega_{p2}) + \omega_{p1}\omega_{p2}(1 + A_{DC}\beta)}$$

2.3.6 Phase Margin

Phase margin At ω

$$\Phi(\omega) = 180 - \sum_{\omega_{p,i}} \tan^{-1}\left(\frac{\omega}{\omega_{p,i}}\right)$$

Unity Gain, Find DC gain in dB, -20db/dec gradient, dec = $\log(\omega/\omega_i)$

2.3.7 Slew Rate

For 2 stages with the 1st stage effective output cap C and saturation current I, and 2nd stage gain A

$$g_{sr} = \frac{I_{1st,sat}}{C_{2nd,in}} A_{2nd}$$

3 Noise

3.1 Noise of different components

Noise at any frequency for...Resistors

$$I_n^2 = \frac{4kT}{R}$$

Mosfets (can be an equivalent V_n) at the gate. With γ usually 2/3

$$I_n^2 = 4kT\gamma g_m \qquad V_n = I_n/g_m$$

Mosfet flicker Noise at frequency f

$$\bar{V}_n^2 = \frac{K}{C_{ox}WL} \times \frac{1}{f} \qquad \bar{I}_{n,1/f}^2 = g_m^2 \bar{V}_n^2$$

Corner Frequency when thernal noise equal flicker noise

$$f_C = \frac{K}{C_{ox}WL} \times \frac{g_m}{4\gamma kT}$$

For Amplifiers, input referred noise voltage must satisfy

$$[V_{n,in}A_V]^2 = V_{n,out}^2 \qquad [V_{n,in}G_m]^2 = I_{n,out}^2$$

The conclusion came from euqating two output current in Degenerated Common Source Stage

$$I_{n,out} = \frac{I_{n,mos}}{Z_s(gm + 1/r_o) + 1} = \frac{g_m V_{n,in}}{Z_s(gm + 1/r_o) + 1}$$

Integrate against df for total noise across BW

4 FeedBack

For a feedback (FB) with open loop gain A_{ol} and feedback factor β. The closed loop gain A_{cl}, let $G = 1 + \beta A_{ol}$, βA_{ol} is called the loop gain.

$$A_{cl} = \frac{A_{ol}}{1 + \beta A_{ol}}$$

Types of Amp's In-Out Impedance and Types of FB can be categorized by (Series Low Z, Shunt High Z)

	Input	Output	Sense	Return
V	HIGH	LOW	Shunt	Series
I	LOW	HIGH	Series	Shunt

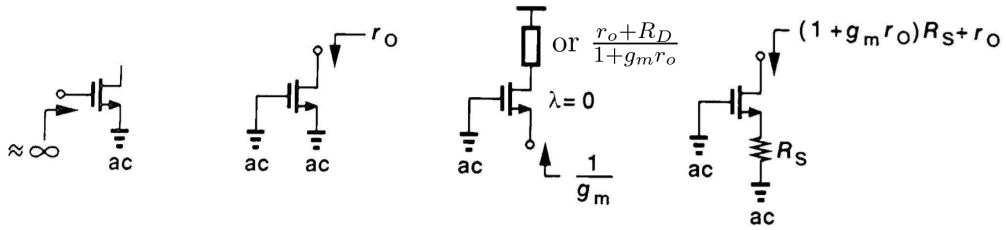
Input Output Impedance of Feedback

R-S	FB	AMP	Z_{in}	Z_{out}
Series-Series	R_0	G_m	$Z_{in}G$	$Z_{out}G$
Series-Shunt	A_I	A_v	$Z_{in}G$	Z_{out}/G
Shunt-Series	A_v	A_I	Z_{in}/G	$Z_{out}G$
Shunt-Shunt	G_m	R_0	Z_{in}/G	Z_{out}/G

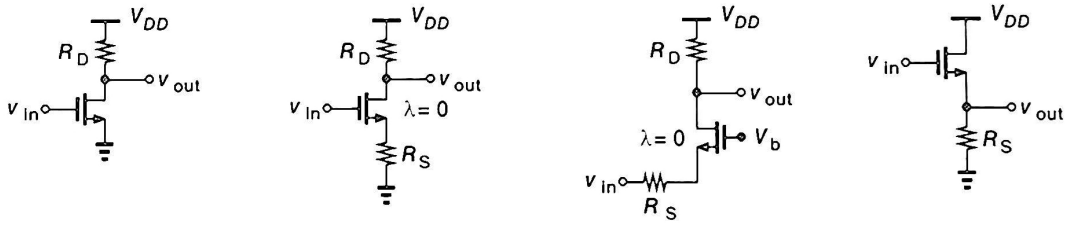
After Breaking feedback, adjust return if sense, adjust sense if return.

	Sense	Return
V	GND	OPEN
I	OPEN	GND

Input and Output Impedances



Voltage Gain Equations



$A_v = -g_m(R_D \| r_O)$

$G_m = g_m$

$A_v = -\frac{R_D}{\frac{1}{g_m} + R_S}$

$G_m = \frac{g_m/R_s}{g_m + (R_s \| r_o)^{-1}} \approx \frac{g_m}{1 + g_m R_s}$

$A_v = \frac{R_D}{\frac{1}{g_m} + R_S}$

$G_m = \frac{-(r_o^{-1} + g_m)}{R_s(r_o^{-1} + g_m) + 1} \approx \frac{-g_m}{1 + g_m R_s}$

$A_v = \frac{R_S \| r_O}{\frac{1}{g_m} + R_S \| r_O}$

$G_m = -g_m$

Feedback Networks

